Supplementary Material

Physics-Data Fusion for Remaining Useful Life Prediction of Electrical Connectors under Salt-Spray Corrosion

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S1 Scope of the Supplementary Evidence

The main article reports the conclusion of the failure-mechanism consistency analysis, whereas this supplementary material presents the detailed experimental evidence supporting that conclusion. The evidence includes the degradation trends of multiple performance parameters, post-test macroscopic appearance, stereomicroscope observations, and SEM/EDS results. It is organized by observation scale and connector region so that the relationships among stress level, corrosion morphology, elemental composition, and functional degradation remain clear.

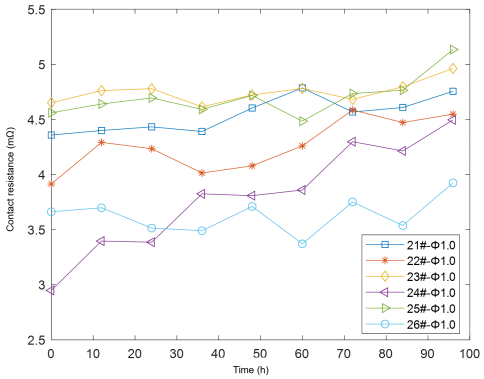
The YB3116F18 and XCE33T41 connectors tested in this study both use gold-plated, copper-based contact components. Under salt-spray exposure, their principal failure driver is electrochemical corrosion involving chloride ions. Because a gold coating naturally contains micropores and local defects, chloride ions can penetrate the coating and form local galvanic cells between the gold layer and the copper substrate. This process promotes oxidation of the copper substrate and the formation of corrosion products. At the same time, the aluminum-alloy shell undergoes oxidation and localized pitting in the chloride-containing environment. As the salt-spray concentration increases, the shell corrosion morphology gradually develops from isolated point-like attack into more extensive surface corrosion.

From the perspective of functional failure, salt-spray-induced degradation of the connectors can be classified into three main categories. First, contact failure occurs when corrosion products accumulate in the contact region and act together with coating damage and local oxidation to reduce the effective conductive area. The contact resistance therefore rises continuously and may eventually exceed its failure threshold. Second, insulation failure occurs when salt crystals and corrosion deposits adhere to the insulating substrate and gradually form continuous conductive paths, causing the insulation resistance to decrease. Third, shell failure occurs after blistering, cracking, and peeling of the shell coating expose the aluminum-alloy substrate to further pitting and surface corrosion, eventually causing the corroded-area ratio to exceed its allowable limit.

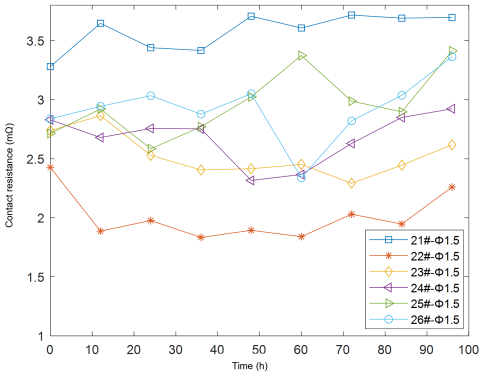
S2 Multi-Parameter Degradation Trends

The degradation behavior changes systematically with salt-spray concentration. At the low concentrations of 0.5% and 1%, corrosion begins mainly as point-like attack at coating micropores, and the performance parameters change comparatively slowly. At the intermediate concentrations of 2% and 3%, local coating damage becomes more pronounced and contact resistance shows a clearer increasing trend. Under the high-concentration conditions of 4% and 5%, coating blistering, cracking, and peeling become more evident, while salt deposits and corrosion products accumulate in larger quantities. Consequently, the probabilities of insulation failure and shell-corrosion failure increase markedly.

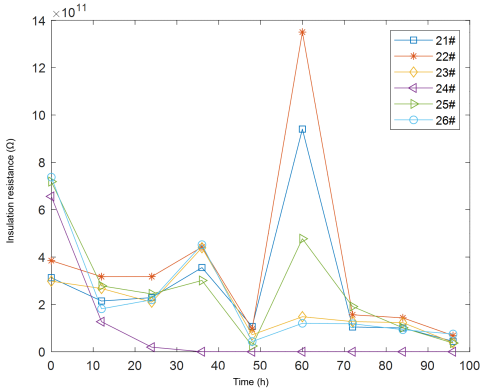
Figure S1 gives representative time histories for the XCE33T41 samples under the 5% salt-spray condition. The contact resistance of both contact groups shows an overall upward trend, although the trajectories fluctuate because of specimen-to-specimen differences and localized corrosion. The insulation resistance generally decreases with exposure time, while the shell corrosion-area ratio increases. The simultaneous evolution of these electrical and structural indicators confirms that salt-spray exposure produces coupled contact, insulation, and shell degradation rather than a single isolated failure mode.



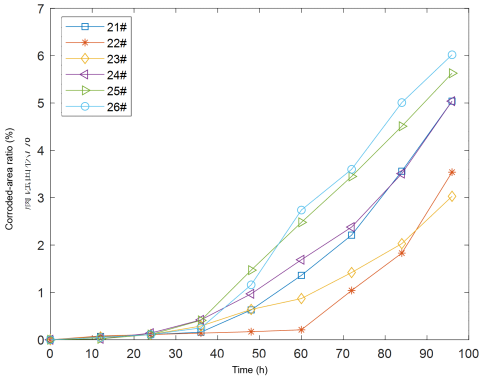
(a) Contact resistance of the Φ1.0 contact group



(b) Contact resistance of the Φ1.5 contact group



(c) Insulation resistance



(d) Shell corrosion-area ratio

Figure S1: Representative degradation trends of key performance parameters under the 5% salt-spray condition.

S3 Macroscopic Appearance after Salt-Spray Testing

The appearance inspection conducted after the salt-spray tests provides macroscopic morphological evidence for the preceding analysis. For the YB3116F18 samples, the photographs illustrate the evolution of corrosion at different salt-spray concentrations; for the XCE33T41 samples, the photographs show representative failure morphologies under the high-concentration condition of 5%. In the YB3116F18 connectors, corrosion is concentrated mainly on the shell and fasteners and is macroscopically manifested as coating discoloration, cracking, and localized peeling. The severity of these features increases progressively with salt-spray concentration. In the XCE33T41 connectors exposed to 5% salt spray, the principal features are discoloration and blistering of the coating in the shell and receptacle regions, accompanied by visible corrosion deposits. The representative macroscopic morphologies are presented in Fig. S2.



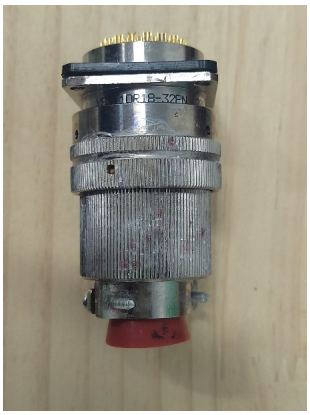
(a) YB3116F18, 0.5%



(b) YB3116F18, 1%



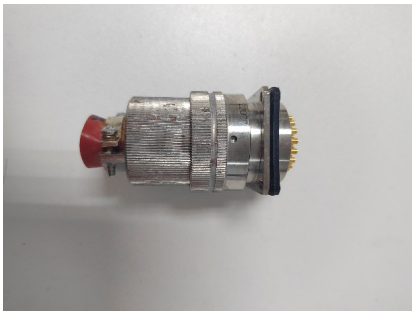
(c) YB3116F18, 2%



(d) YB3116F18, 3%



(e) YB3116F18, 4%



(f) YB3116F18, 5%



(g) XCE33T41, 5%, overall view

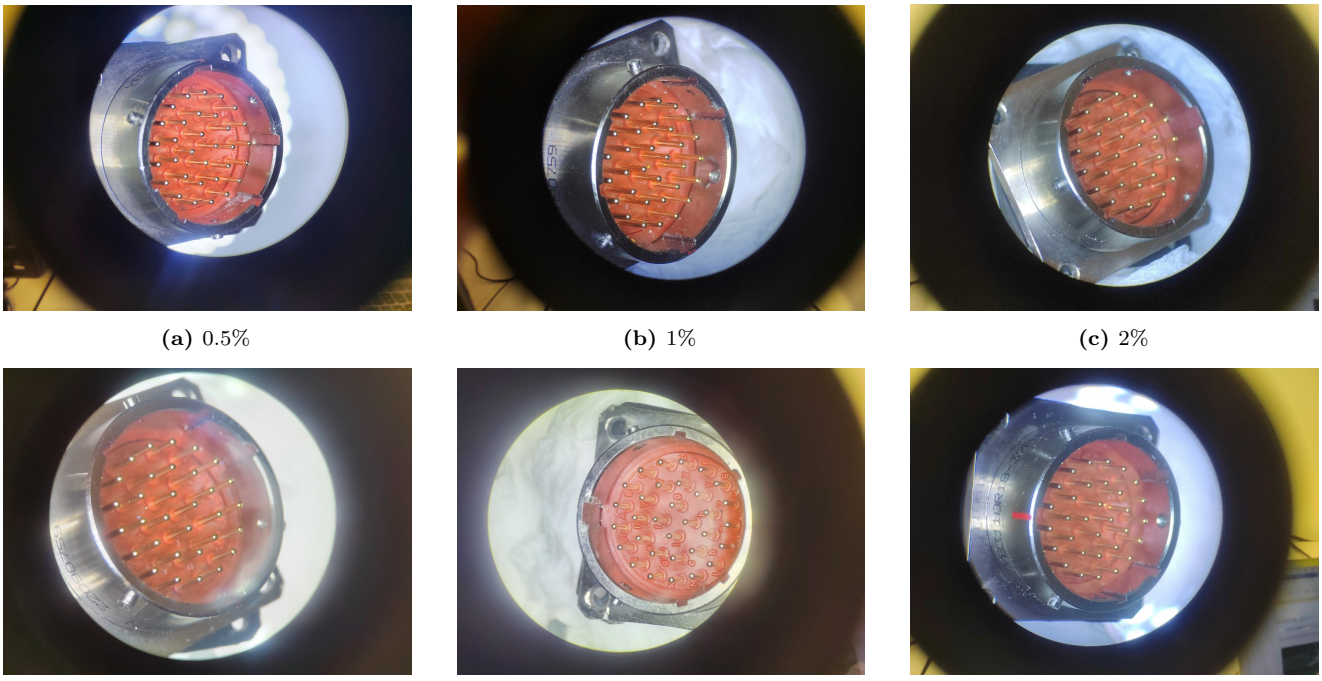


(h) XCE33T41, 5%, end-region view

Figure S2: Representative macroscopic appearance of electrical connector samples after salt-spray testing.

S4 Stereomicroscope Observations

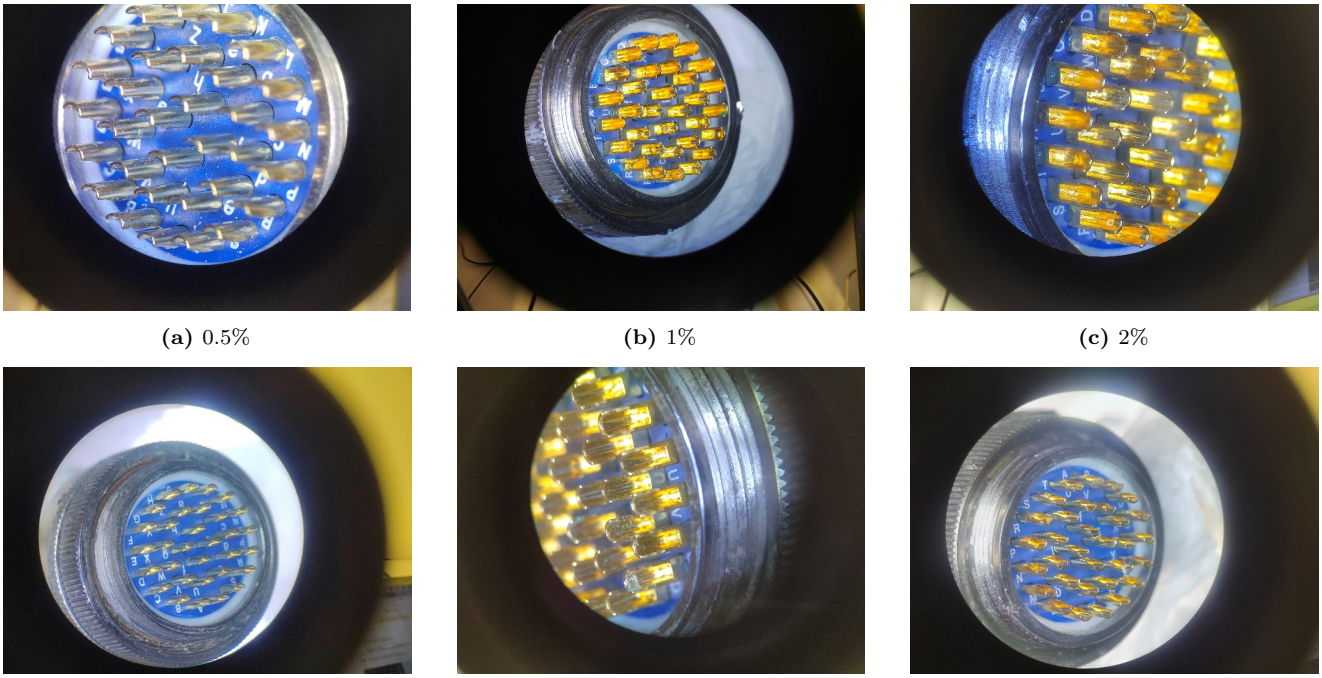
Further stereomicroscope inspection shows gray-white corrosion products, coating blistering, and coating peeling on the YB3116F18 samples exposed to different salt-spray concentrations. Salt precipitates adhere to the contact-pair surfaces, solder-cup contact surfaces, and shell regions, and these corrosion features become progressively more pronounced as the concentration increases. The three regions exhibit different local manifestations: the contact-pair and solder-cup surfaces show localized deposits and changes in metallic appearance, whereas the shell shows broader coating discoloration, blistering, and peeling. Under the 5% salt-spray condition, the XCE33T41 samples likewise exhibit corrosion deposits in the shell and receptacle regions, and local bulging of the material is visible at the bottom of the solder cup. These results show that both connector types follow a degradation sequence from surface deposition and coating damage to the expansion of localized corrosion. The corresponding observations are provided in Figs. S3–S6.



(a) 0.5% **(b)** 1% **(c)** 2%

(d) 3% **(e)** 4% **(f)** 5%

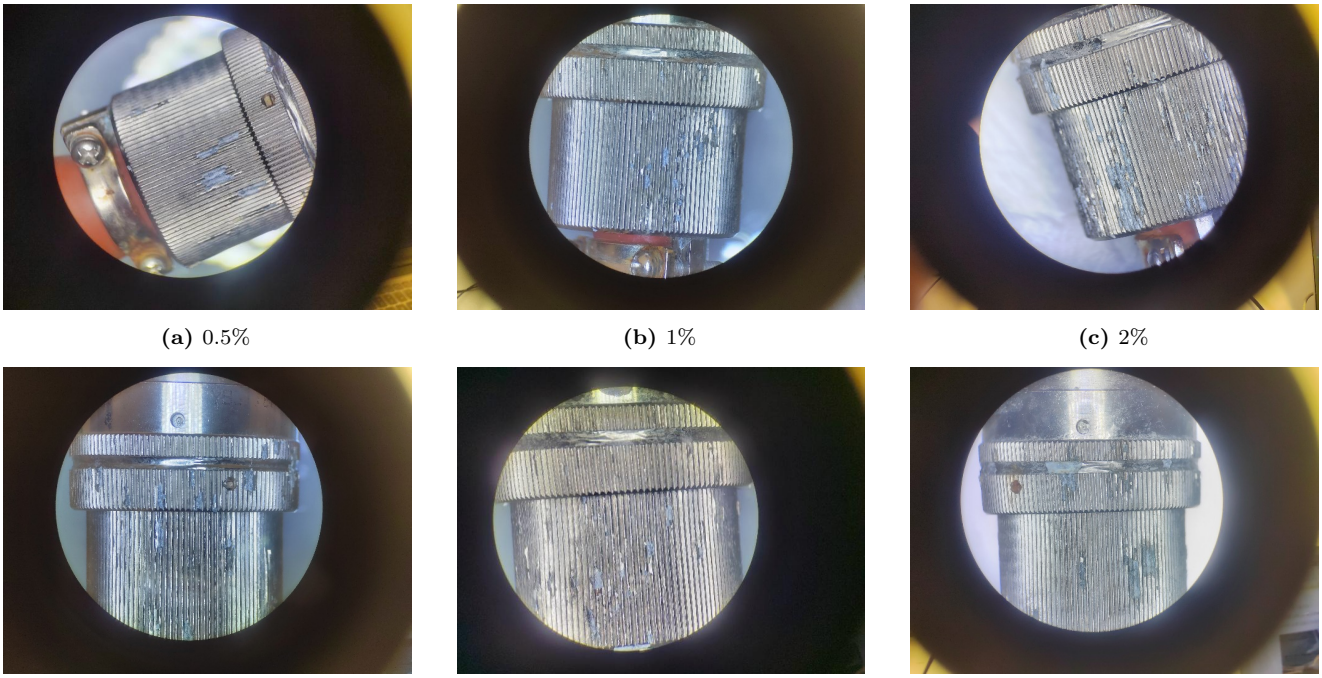
Figure S3: Stereomicroscope images of the YB3116F18 contact-pair region after exposure to different salt-spray concentrations.



(a) 0.5% **(b)** 1% **(c)** 2%

(d) 3% **(e)** 4% **(f)** 5%

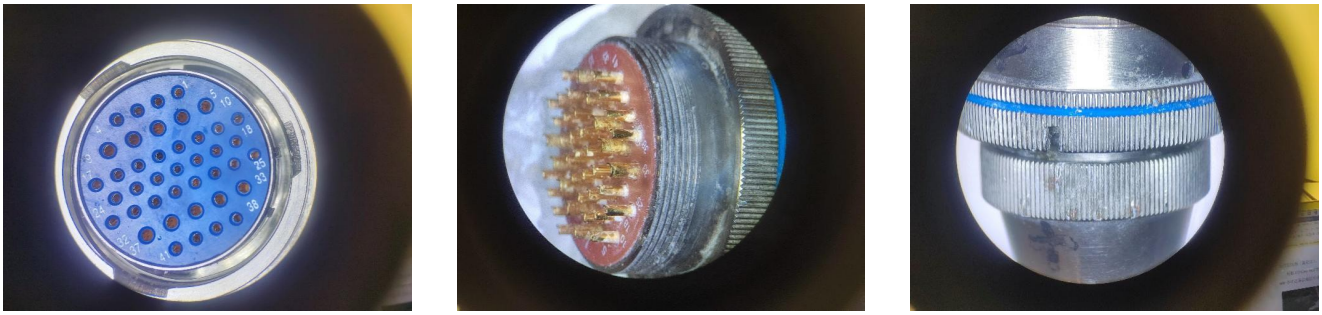
Figure S4: Stereomicroscope images of the YB3116F18 solder-cup region after exposure to different salt-spray concentrations.



(a) 0.5% **(b)** 1% **(c)** 2%

(d) 3% **(e)** 4% **(f)** 5%

Figure S5: Stereomicroscope images of the YB3116F18 shell after exposure to different salt-spray concentrations.



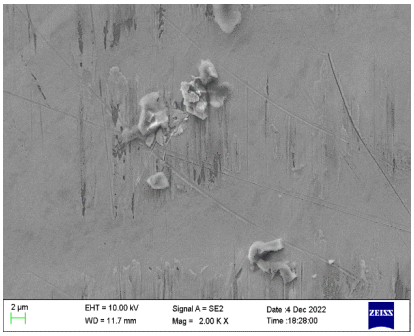
(a) Contact-pair region **(b)** Solder-cup region **(c)** Shell region

Figure S6: Representative stereomicroscope images of the XCE33T41 connector after exposure to the 5% salt-spray condition.

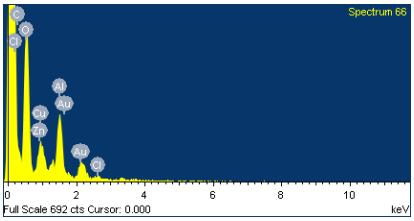
S5 SEM and EDS Evidence

SEM and EDS analyses were performed to further reveal the microscopic failure mechanism of salt-spray corrosion. To provide representative evidence without excessive duplication, the contact-region morphologies of YB3116F18 samples at several salt-spray concentrations are shown together with complementary high-stress results from XCE33T41 samples exposed to 5% salt spray. The SEM images show that, as the salt-spray concentration increases, corrosion of the contact surface develops from slight localized deposition into more extensive corrosion and surface-layer damage. Corrosion products accumulate more heavily, and some regions show clearer evidence of coating damage and exposure of the underlying material.

The EDS results provide complementary compositional evidence, although the detected elements vary among sampling locations because EDS probes a small local area. The YB3116F18 contact-region spectra repeatedly contain oxygen and material-related elements such as copper or aluminum, and chloride is detected at several representative sites. Their co-occurrence supports the participation of chloride in the continuing corrosion process and the oxidation of copper-bearing or aluminum-bearing materials at damaged or contaminated regions. In mechanistic terms, coating defects and micropores on the contact surface preferentially become initiation sites for localized corrosion. As corrosion products continue to accumulate, the effective conductive area of the interface decreases and the resistance of the interfacial film increases, thereby producing the sustained increase in contact resistance observed during the degradation tests.



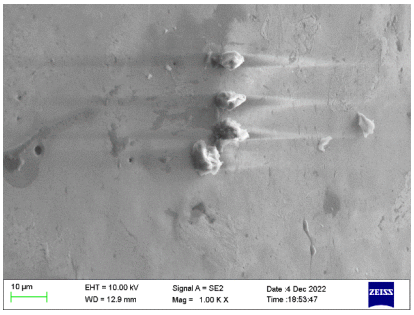
(a) 0.5%, SEM



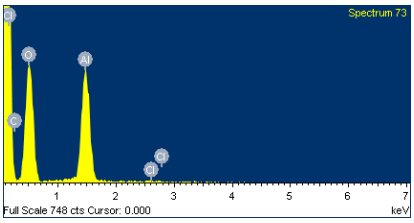
(b) 0.5%, EDS spectrum

Element	Weight%	Atomic%
C K	19.18	31.34
O K	45.56	55.89
Al K	9.55	6.94
Cl K	0.80	0.44
Cu L	10.66	3.29
Zn L	3.38	1.02
Au M	10.87	1.08
Totals	100.00	

(c) 0.5%, composition



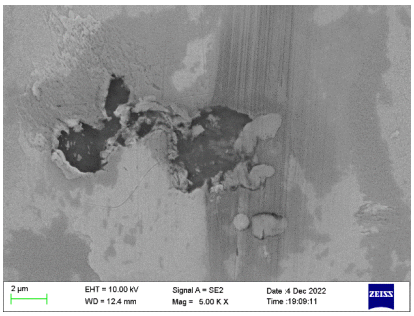
(d) 1%, SEM



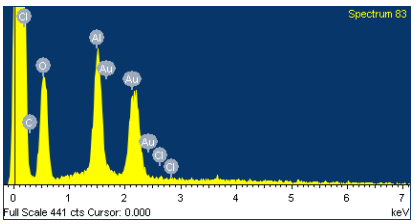
(e) 1%, EDS spectrum

Element	Weight%	Atomic%
C K	15.31	22.12
O K	53.42	57.93
Al K	30.26	19.46
Cl K	1.01	0.49
Totals	100.00	

(f) 1%, composition



(g) 2%, SEM

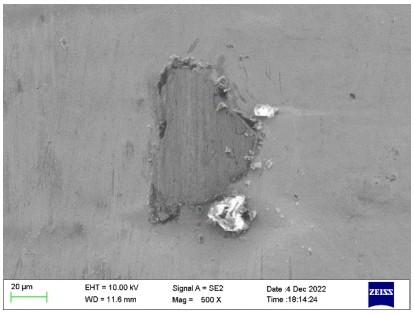


(h) 2%, EDS spectrum

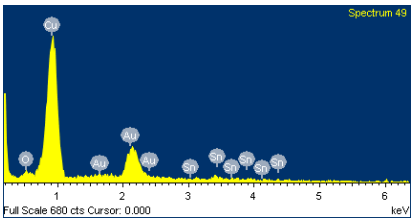
Element	Weight%	Atomic%
C K	6.28	15.67
O K	32.21	60.31
Al K	15.02	16.68
Cl K	0.40	0.33
Au M	46.09	7.01
Totals	100.00	

(i) 2%, composition

Figure S7: SEM and EDS results for the YB3116F18 contact region at 0.5%–2% salt-spray concentrations.



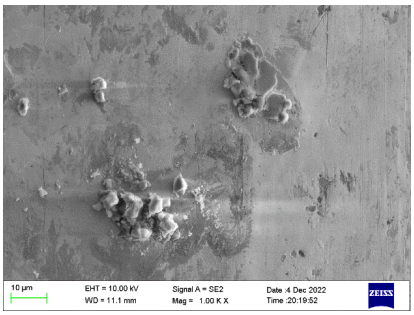
(a) 3%, SEM



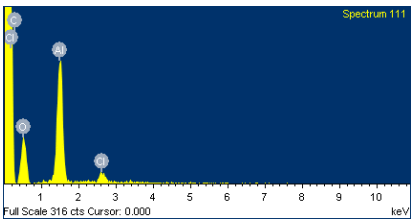
(b) 3%, EDS spectrum

Element	Weight%	Atomic%
O K	1.71	8.04
Cu L	65.96	78.05
Sn L	6.23	3.95
Au M	26.09	9.96
Totals	100.00	

(c) 3%, composition



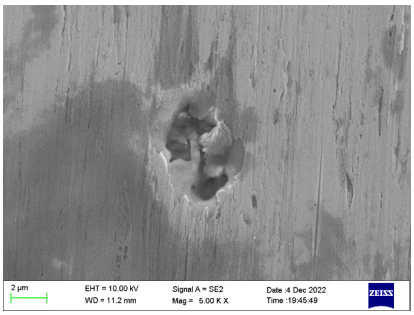
(d) 4%, SEM



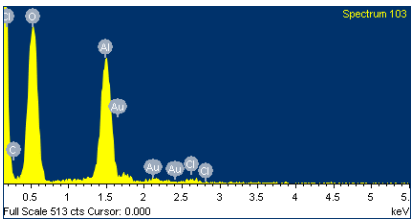
(e) 4%, EDS spectrum

Element	Weight%	Atomic%
C K	7.15	12.13
O K	37.16	47.32
Al K	47.39	35.78
Cl K	8.30	4.77
Totals	100.00	

(f) 4%, composition



(g) 5%, SEM



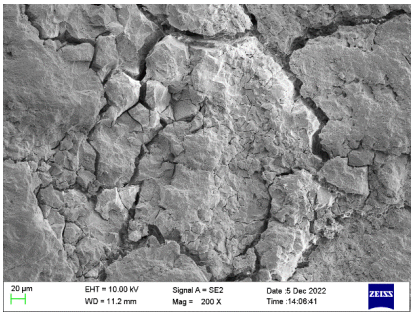
(h) 5%, EDS spectrum

Element	Weight%	Atomic%
C K	11.51	16.93
O K	59.71	65.95
Al K	24.63	16.13
Cl K	1.50	0.75
Au M	2.64	0.24
Totals	100.00	

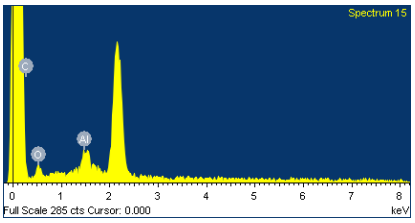
(i) 5%, composition

Figure S8: SEM and EDS results for the YB3116F18 contact region at 3%–5% salt-spray concentrations.

The YB3116F18 shell surfaces show cracked, layered, and locally detached corrosion products, and the surface damage generally becomes more extensive at higher salt-spray concentrations. The shell spectra are dominated by carbon, oxygen, and aluminum, with nickel or sodium detected at some locations. Chloride is not detected uniformly in every shell spectrum because the EDS measurement represents only the selected microscopic area. Nevertheless, the shell morphology together with the oxygen-rich composition is consistent with oxidation of the aluminum-alloy shell, progressive coating degradation, and local exposure of shell-related materials in the salt-spray environment.



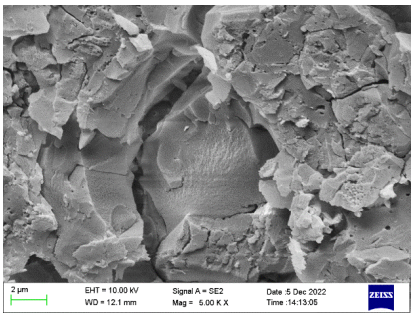
(a) 0.5%, SEM



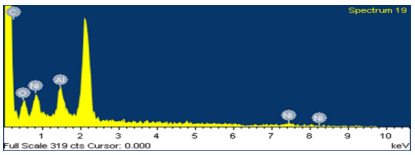
(b) 0.5%, EDS spectrum

Element	Weight%	Atomic%
C K	39.82	51.18
O K	36.64	35.35
Al K	23.54	13.47
Totals	100.00	

(c) 0.5%, composition



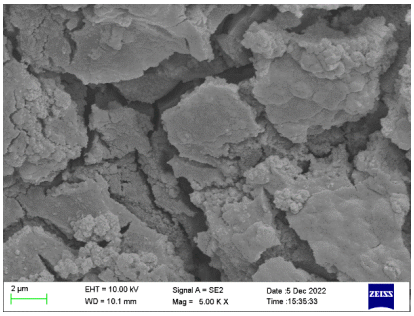
(d) 1%, SEM



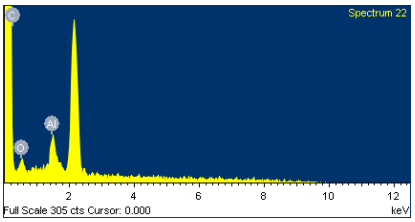
(e) 1%, EDS spectrum

Element	Weight%	Atomic%
C K	27.24	47.82
O K	20.13	26.53
Al K	15.97	12.48
Ni L	36.66	13.17
Totals	100.00	

(f) 1%, composition



(g) 2%, SEM



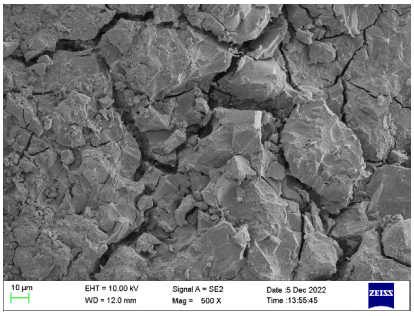
(h) 2%, EDS spectrum

Element	Weight%	Atomic%
C K	30.84	42.22
O K	37.33	38.37
Al K	31.84	19.41
Totals	100.00	

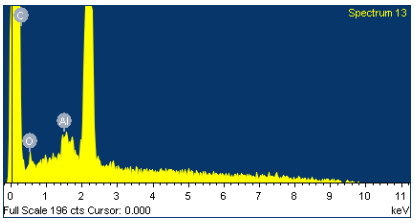
(i) 2%, composition

Figure S9: SEM and EDS results for the YB3116F18 shell at 0.5%–2% salt-spray concentrations.

The XCE33T41 results under the 5% condition provide an additional high-stress comparison. In the contact region, the SEM image shows substantial surface deposits and local damage, while the EDS result detects oxygen, aluminum, copper, chloride, and gold among the principal material-related elements at the selected site. This composition is consistent with chloride-assisted electrochemical corrosion involving the contact coating and copper-based substrate. In the shell region, the corrosion products are rich in carbon, oxygen, and aluminum and also contain component- or deposit-related elements at the selected point. Together, the XCE33T41 observations reproduce the same progression from deposition and coating deterioration to localized corrosion that is observed in the YB3116F18 samples.



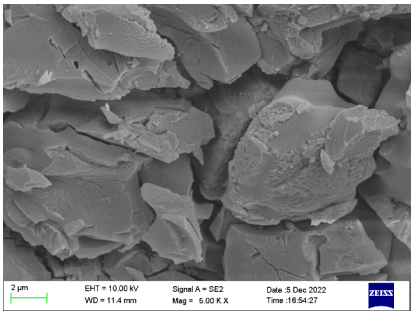
(a) 3%, SEM



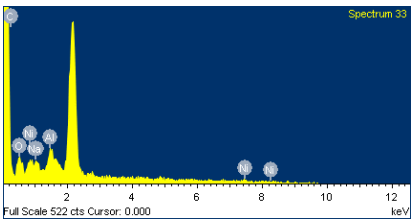
(b) 3%, EDS spectrum

Element	Weight%	Atomic%
C K	30.29	43.87
O K	25.27	27.48
Al K	44.43	28.65
Totals	100.00	

(c) 3%, composition



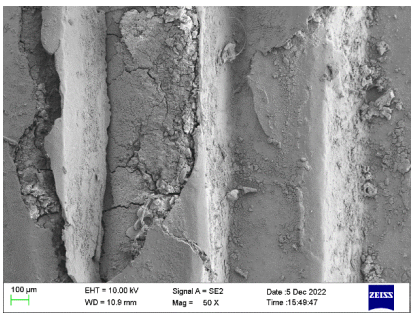
(d) 4%, SEM



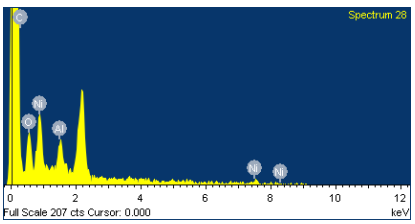
(e) 4%, EDS spectrum

Element	Weight%	Atomic%
C K	27.00	44.76
O K	23.03	28.65
Na K	10.84	9.39
Al K	9.86	7.27
Ni L	29.27	9.92
Totals	100.00	

(f) 4%, composition



(g) 5%, SEM

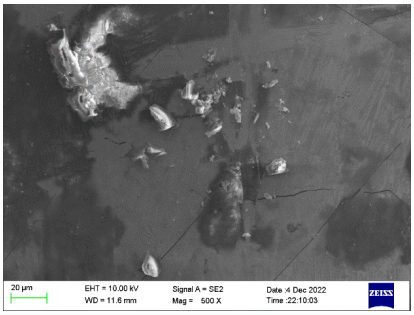


(h) 5%, EDS spectrum

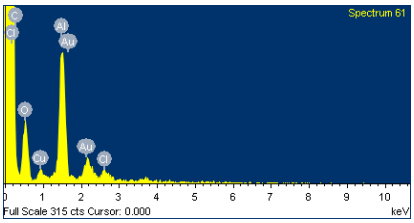
Element	Weight%	Atomic%
C K	22.54	43.53
O K	20.92	30.32
Al K	8.23	7.07
Ni L	48.31	19.08
Totals	100.00	

(i) 5%, composition

Figure S10: SEM and EDS results for the YB3116F18 shell at 3%–5% salt-spray concentrations.



(a) SEM morphology

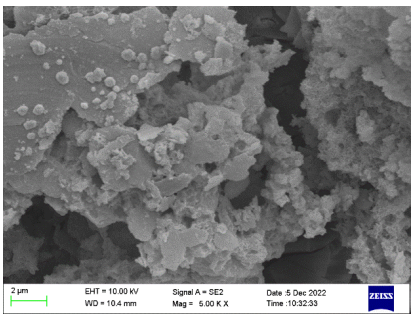


(b) EDS spectrum

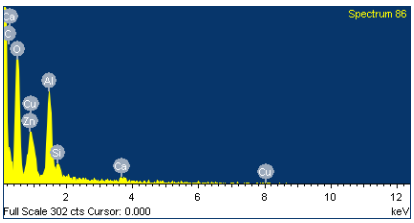
Element	Weight%	Atomic%
C K	17.83	31.59
O K	30.40	40.42
Al K	28.76	22.67
Cl K	3.95	2.37
Cu L	3.94	1.32
Au M	15.12	1.63
Totals	100.00	

(c) Elemental composition

Figure S11: Representative SEM and EDS results for the XCE33T41 contact region after exposure to the 5% salt-spray condition.



(a) SEM morphology



(b) EDS spectrum

Element	Weight%	Atomic%
C K	15.73	25.49
O K	44.84	54.57
Al K	14.84	10.71
Si K	3.31	2.29
Ca K	2.56	1.24
Cu L	14.04	4.30
Zn L	4.69	1.40
Totals	100.00	

(c) Elemental composition

Figure S12: Representative SEM and EDS results for the XCE33T41 shell region after exposure to the 5% salt-spray condition.

S6 Mechanism Synthesis

The multi-scale evidence supports a consistent salt-spray corrosion mechanism for the two connector types. Macroscopic and stereomicroscope observations establish the progression from surface deposition and discoloration to coating damage and localized corrosion. SEM images further reveal corrosion-product accumulation, pits, cracks, and local surface-layer destruction. EDS confirms oxygen-rich corrosion products and detects chloride at representative contact-region sites, together with copper-, aluminum-, gold-, nickel-, or other material-related elements according to the sampled component and location.

These observations are consistent with the degradation indicators used in the main article. Corrosion-product accumulation and local coating damage reduce the effective conductive area and increase contact-interface resistance, explaining the progressive contact-resistance degradation. Deposits on insulating surfaces can promote leakage paths and insulation-resistance reduction, while shell oxidation and coating failure explain the increase in corrosion-area ratio. Although the severity and local morphology vary among samples and stress levels, the dominant electrochemical-corrosion pathway remains consistent across the tested conditions. This consistency supports the cross-stress lifetime-equivalence assumption used by the physics-data fusion RUL framework.